

Serverless URL Shortener with Analytics

Cloud Computing Project using AWS



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Serverless Core

Short links, instant redirects, click analytics, and automatic cloud scalability.

Lambda + API Gateway + DynamoDB

S3 + CloudFront frontend

What is a URL Shortener?

A URL shortener converts long web addresses into compact links that are easier to share and track.

Original URL

<https://amazon.in/products/mobile/iphone>



Short URL

<https://short.ly/aX92k>



MAIN FEATURES

- Short URL generation
- Fast redirection
- Click analytics
- Cloud deployment

Objective

Build a scalable cloud-based URL shortener using AWS serverless services.



Application Goal

Use cloud-native architecture to reduce long URLs, process redirects, and track user click activity.

0 servers

Event-driven Lambda backend.

Auto scale

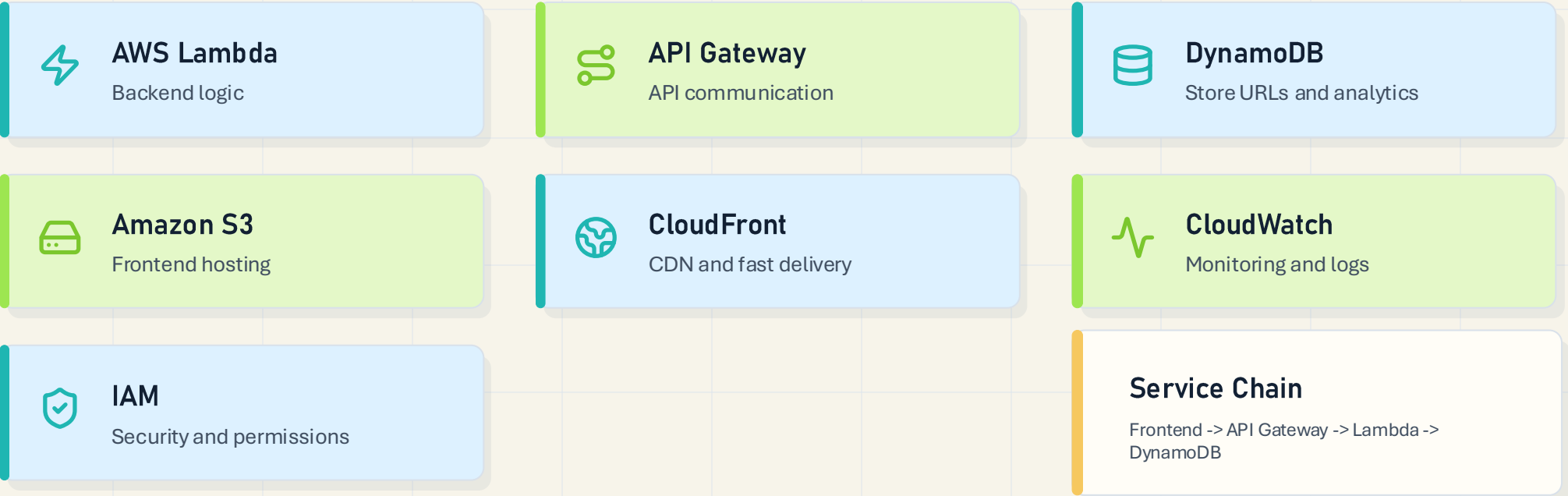
Capacity grows with request traffic.

Goals

- Reduce long URLs
- Track user clicks
- Minimize server management

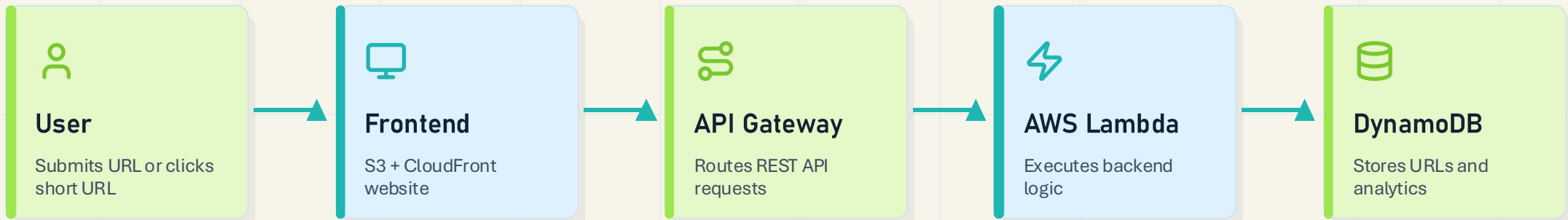
AWS Services Used

Each service handles a focused part of the serverless URL shortener workflow.



System Architecture

A serverless flow handles link creation, redirection, and analytics without managing servers.



ANALYTICS FLOW

User click -> Lambda runs -> Click count updated -> Analytics stored

Frontend and Backend

The frontend is hosted statically while Lambda handles backend execution only when needed.



Frontend

Amazon S3

CloudFront CDN

- Fast website loading
- Low hosting cost
- Global content delivery



Backend

AWS Lambda Functions

- Create short URL
- Redirect URLs
- Update analytics
- Auto scaling execution

Database and Analytics

DynamoDB stores URL mappings and click data for fast lookup and scalable analytics.



DynamoDB Stores

- Original URL
- Short ID
- Click count
- Timestamp

High-performance NoSQL



Analytics Tracks

- Total clicks
- Popular URLs
- User activity
- Redirection events

Scalable analytics layer

Security and Monitoring

IAM protects AWS resources while CloudWatch gives visibility into application behavior.



IAM

Identity and Access Management

- Access control
- Secure permissions
- Protect AWS resources



CloudWatch

Logs, metrics, and performance visibility

- Monitoring logs
- Error tracking
- Performance analysis

Advantages and Future Scope

The project is lightweight today and can grow into a richer analytics product.



Advantages

- Fully serverless
- Cost efficient
- Highly scalable
- Easy deployment
- Low maintenance



Future Scope

- QR code support
- Custom URLs
- User authentication
- Geographic analytics
- Admin dashboard

NEXT ITERATION: richer dashboards + user accounts

Conclusion



AWS cloud services can be combined to build scalable and efficient serverless applications.

Project Summary

The system shortens URLs, redirects users quickly, and records analytics using AWS serverless components.

Serverless + Scalable + Observable

Key Learnings

- Cloud computing concepts
- Serverless architecture
- AWS deployment
- Monitoring and scalability